



Lista 5

$$1-a) \vec{BF} = \vec{AF} - \vec{AB} = \vec{f} - \vec{b}$$

$$b) \vec{AG} = \vec{AC} + \vec{AF} - \vec{AB} = \vec{c} + \vec{f} - \vec{b}$$

$$c) \vec{AE} = \vec{AF} - \vec{AB} = \vec{f} - \vec{b}$$

$$d) \vec{BG} = \vec{AF} - \vec{AB} + \vec{AC} - \vec{AB} = \vec{f} - \vec{b} + \vec{c} - \vec{b} = \vec{f} + \vec{c} - 2\vec{b}$$

$$e) \vec{HD} = \vec{HA} + \vec{AB} = -\vec{f} + \vec{b}$$

$$f) \vec{AD} + \vec{FG} = -\vec{b} + \vec{c} - \vec{b} = \vec{c}$$

$$g) \vec{AD} + \vec{HG} = \vec{AC} - \vec{AB} + \vec{AB} = \vec{c}$$

$$h) \vec{HE} + \vec{AG} - \vec{EF} = 2\vec{c} + \vec{f} - 2\vec{b}$$

$$i) 2\vec{AD} - \vec{FG} - \vec{BH} + \vec{GH} = 2\vec{c} - 2\vec{b} - \vec{c} + \vec{b} - \vec{b} + \vec{b} + (\vec{c} - \vec{b}) = 2\vec{c} - 4\vec{b} + \vec{f}$$

$$2a) \vec{DF} = 2\vec{DE} + \vec{DC}$$

$$e) \vec{EC} = 2\vec{DE} - \vec{DC}$$

$$b) \vec{DA} = 2\vec{DE} + 2\vec{DC}$$

$$f) \vec{EB} = \vec{DE} - (2\vec{DC} - \vec{DE})$$

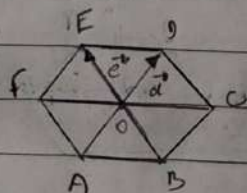
$$c) \vec{DB} = 2\vec{DC} + \vec{DE}$$

$$g) \vec{OB} = \vec{DC} + \vec{DE} - (2\vec{DC} + \vec{DE})$$

$$d) \vec{DO} = \vec{DC} + \vec{DE}$$

$$h) \vec{AF} = 2\vec{DE} + 2\vec{DC} - (2\vec{DE} + \vec{DC})$$

$$3-a) \vec{OA} + \vec{OB} + \vec{OC} + \vec{OD} + \vec{OE} + \vec{OF} \\ = -\vec{d} + (-\vec{e}) + (\vec{d} + (-\vec{e})) + \vec{d} + \vec{e} + (\vec{e} - \vec{d}) \\ = -\vec{d} - \vec{e} + \vec{d} - \vec{e} + \vec{d} + \vec{e} + \vec{e} - \vec{d} \\ = \vec{0}$$



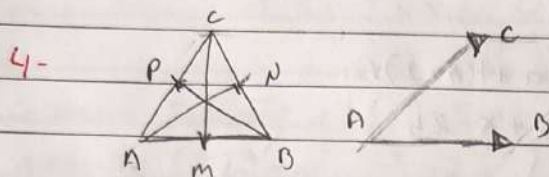
$$5) \vec{AB} + \vec{BC} + \vec{CD} + \vec{DE} + \vec{EF} + \vec{FA} = \vec{AA} = \vec{0} \quad (\text{caminho fechado})$$

$$c) \vec{AB} + \vec{BC} + \vec{CD} + \vec{DE} + \vec{EF} = \vec{AF} = -\vec{d} - (-\vec{d} + \vec{e}) = -\vec{e}$$

$$d) \vec{OA} + \vec{OB} + \vec{OD} + \vec{OE} = -\vec{d} + (-\vec{e}) + \vec{d} + \vec{e} = \vec{0}$$

$$e) \vec{OC} + \vec{AF} + \vec{EF} = (\vec{d} - \vec{e}) + (-\vec{d} + (-\vec{d} + \vec{e})) + \vec{e} + (-\vec{d} + \vec{e}) = 2\vec{e} - 2\vec{d}$$

$$f) \vec{AF} + \vec{DE} = (-\vec{d} + (-\vec{d} + \vec{e})) + \vec{d} - \vec{e} = -\vec{d}$$



$$\vec{AN} = (\vec{AC} + \vec{AB}) \div 2 = \frac{\vec{AB}}{2} + \frac{\vec{AC}}{2}$$

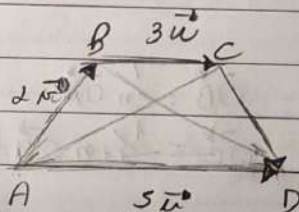
$$\vec{BP} = -\vec{AB} + \frac{\vec{AC}}{2} = \frac{\vec{AC}}{2} - \vec{AB}$$

$$\vec{CM} = -\vec{AC} + \frac{\vec{AB}}{2} = \frac{\vec{AB}}{2} - \vec{AC}$$

$$5-a) \vec{CD} = \vec{AB} + \vec{BC} - \vec{AD}$$

$$2\vec{n} + 3\vec{u} - 5\vec{u}$$

$$2\vec{n} - 2\vec{u}$$



$$\vec{BD} = \vec{BA} + \vec{AD} = -2\vec{n} + 5\vec{u}$$

$$\vec{CA} = \vec{CB} + \vec{BA} = -3\vec{u} - 2\vec{n}$$

b) $\vec{BC}$ e $\vec{AD}$ são paralelos, pois

$\vec{AD} = \frac{5}{3}\vec{BC}$; dessa forma $\vec{AD}$ é múltiplo, então paralelos

Assim ABCD é um trapézio escaleno.



6- Sabendo, $\vec{OA} = \vec{a}$, $\vec{OB} = \vec{b}$, $\vec{OC} = \vec{c}$, $\vec{AD} = \frac{1}{4}\vec{c}$ e $\vec{BE} = \frac{5}{6}\vec{a}$
 $\vec{DE} = ?$

$$I \vec{AB} = -\vec{OA} + \vec{OB} = -\vec{a} + \vec{b} \quad , \quad \vec{DE} = \vec{AB} - \vec{DA} + \vec{BE}$$

$$II \vec{AB} - \vec{AD} = \vec{DB} = \vec{a} + \vec{b} - \frac{1}{4}\vec{c} \quad -\vec{a} + \vec{b} - \frac{1}{4}\vec{c} + \frac{5}{6}\vec{a}$$

$$-\frac{1}{6}\vec{a} + \vec{b} - \frac{1}{4}\vec{c}$$

7- Sabendo, $\vec{OA} = \vec{a} + 2\vec{b}$, $\vec{OB} = 3\vec{a} + 2\vec{b}$ e $\vec{OC} = 5\vec{a} + x\vec{b}$, $x?$ para LD

$$\vec{AC} = -\vec{OA} + \vec{OC} = 5\vec{a} - \vec{a} + x\vec{b} - 2\vec{b} = 4\vec{a} + (x-2)\vec{b}$$

$$\vec{BC} = -\vec{OB} + \vec{OC} = 5\vec{a} - 3\vec{a} + x\vec{b} - 2\vec{b} = 2\vec{a} + (x-2)\vec{b}$$

Para $\vec{AC}$ e $\vec{BC}$ LD se $K \neq 0$, tal que $\vec{AC} = K \cdot \vec{BC}$

$$\vec{AC} = K \cdot \vec{BC}$$

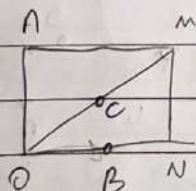
$$4\vec{a} + (x-2)\vec{b} = K \cdot 2\vec{a} + (x-2)\vec{b} \quad x-2 = K \cdot (x-2)$$

$$4 = 2K \quad x-2 = 2 \cdot (x-2)$$

$$K = 2 \quad (L.D) \quad x-2 = 2x-4$$

$$x = 2-2$$

8-



$$\vec{OB} = \frac{1}{n}\vec{ON}$$

$A, B, C \rightarrow$ colineares

$$\vec{OC} = \frac{1}{1+n}\vec{OM} \rightarrow \frac{1}{1+n}(\vec{OA} + \vec{ON})$$

$$\vec{AB} \text{ e } \vec{AC} : \vec{AB} = \vec{OB} - \vec{OA}$$

$$\vec{AC} = \vec{OC} - \vec{OA}$$

$$\vec{AB} = \frac{1}{n}\vec{ON} - \vec{OA}$$

$$\vec{AC} = \frac{1}{1+n}\vec{OA} + \frac{1}{1+n}\vec{ON} - \vec{OA}$$

$$\vec{AC} = \frac{1}{1+n}\vec{ON} + \left(\frac{1}{1+n} - 1\right)\vec{OA}$$

$$\vec{AC} = \frac{1}{1+n}\vec{ON} - \frac{n}{1+n}\vec{OA}$$

$$\text{Relação: } \vec{AC} = K \cdot \vec{AB}$$

$$\frac{1}{1+n}\vec{ON} - \frac{n}{1+n}\vec{OA} = K \cdot \left(\frac{1}{n}\vec{ON} - \vec{OA}\right)$$

$$\frac{1}{1+n}\vec{ON} - \frac{n}{1+n}\vec{OA} = \frac{K}{n}\vec{ON} - K\vec{OA}$$

$$\frac{1}{1+n} = \frac{K}{n}$$

$$-\frac{n}{1+n} = -K$$

$$K = \frac{n}{1+n}$$

São paralelos

$\therefore \vec{AC}$ e $\vec{AB}$ são paralelos e compartilham o ponto A, assim estão na mesma reta, ou seja, são colineares

FORONI



9- $\vec{u}$ e $\vec{v}$ são para para um plano e formam L.I. para $\{2\vec{u} + \vec{v}, \vec{u} - 2\vec{v}\}$ L.I., temos:

$$k_1(2\vec{u} + \vec{v}) + k_2(\vec{u} - 2\vec{v}) = \vec{0}$$

$$2k_1\vec{u} + k_1\vec{v} + k_2\vec{u} - 2k_2\vec{v} = \vec{0}$$

$$(2k_1 + k_2)\vec{u} + (k_1 - 2k_2)\vec{v} = \vec{0}$$

$$\begin{cases} 2k_1 + k_2 = 0 \rightarrow 2 \cdot 2k_2 + k_2 = 0 \Rightarrow 5k_2 = 0 \\ k_1 - 2k_2 = 0 \rightarrow k_1 = 2k_2 \end{cases} \quad k_2 = 0$$

$$k_1 = 2 \cdot 0$$

$\therefore$ única solução é $k_1 = k_2 = 0$,

portanto $\{2\vec{u} + \vec{v}, \vec{u} - 2\vec{v}\}$ é L.I., assim

formam uma base para o plano

10-a) $\vec{u} + \vec{v}$, $\vec{u} - \vec{v} + \vec{w}$ e $\vec{u} + \vec{v} + \vec{w}$ são L.I.

$$(k_1 + k_2 + k_3) + (k_1 - k_2 + k_3)(k_2 + k_3)$$

$$\begin{cases} k_1 + k_2 + k_3 = 0 \rightarrow k_1 + 0 = 0 \\ k_1 - k_2 + k_3 = 0 \rightarrow k_1 = 0 \\ k_2 + k_3 = 0 \Rightarrow k_3 = -k_2 = 0 \end{cases}$$

$\therefore$ Os vetores são L.I.; $k_1 = k_2 = k_3 = 0$

b) $t = a\vec{u} + b\vec{v} + c\vec{w}$; $\vec{u} + \vec{t}$, $\vec{v} + \vec{t}$ e $\vec{w} + \vec{t}$

$$V_1 = \vec{u} + \vec{t} = (1+a)\vec{u} + b\vec{v} + c\vec{w}$$

$$V_2 = \vec{v} + \vec{t} = a\vec{u} + (1+b)\vec{v} + c\vec{w}$$

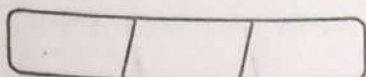
$$V_3 = \vec{w} + \vec{t} = a\vec{u} + b\vec{v} + (1+c)\vec{w}$$

$$M = \begin{pmatrix} 1+a & b & c \\ c & 1+b & c \\ a & b & 1+c \end{pmatrix} = 1+a+b+c$$

para L.I. $\det \neq 0$, logo

$$1+a+b+c \neq 0$$

$$a+b+c \neq -1$$



$$11-a) \vec{AB} = B - A = (1-1, 0-3, -1-2) = (0, -3, -3)$$

$$\vec{BC} = C - B = (1-1, 1-0, 0-(-1)) = (0, 1, 1)$$

$$\vec{CA} = A - C = (1-1, 3-1, 2-0) = (0, 2, 2)$$

$$b) \vec{AB} + \frac{2}{3} \vec{BC} = (0, -3, -3) + (0, \frac{2}{3}, \frac{2}{3})$$
$$(0+0, -3+\frac{2}{3}, -3+\frac{2}{3})$$
$$(0, -\frac{7}{3}, -\frac{8}{3})$$

$$c) C + \frac{1}{2} \vec{AB} = (1, 1, 0) + (0, -\frac{3}{2}, -\frac{3}{2})$$
$$(1+0, 1+(-\frac{3}{2}), 0+(-\frac{3}{2}))$$
$$(1, -\frac{1}{2}, -\frac{3}{2})$$

$$d) A - 2 \vec{BC} = (1, 3, 2) - (0, 2, 2)$$
$$(1-0, 3-2, 2-2)$$
$$(1, 1, 0)$$

$$12-a) \begin{pmatrix} 2 & 3 \\ 0 & 2 \end{pmatrix} \quad 0+4=+4 \neq 0 \quad \text{LI}$$

$$b) \begin{pmatrix} 3 & 0 \\ -2 & 0 \end{pmatrix} \quad 0+0 \cdot 0=0 \quad \text{N\~ao e LI (L.D)}$$

$$c) \begin{cases} 2K_1 = 0 & K_1 = 0 & K_1 = K_2 = 0 \\ 3K_1 + 3K_2 = 0 & 3 \cdot 0 + 3 \cdot K_2 = 0 & \text{S\~ao LI} \\ 4K_1 + 3K_2 = 0 & 4 \cdot 0 = 0 & \end{cases}$$

$$d) \begin{pmatrix} 1 & 1 & 1 \\ -1 & 1 & 1 \\ 2 & 0 & 1 \end{pmatrix} = -2 + 0 + 1 + 1 + 0 + 2 = 2 \neq 0 \quad \text{e LI}$$

$$e) \begin{pmatrix} 1 & -1 & -1 \\ -1 & 2 & 2 \\ 1 & 1 & 2 \end{pmatrix} = 4 + 1 - 2 + 2 - 2 - 2 = 1 \neq 0 \quad \text{e LI}$$



f) $\begin{pmatrix} 1 & 0 & 2 \\ 0 & 0 & 0 \\ 1 & 1 & 3 \end{pmatrix} \rightarrow \Delta \text{Det} = 0 : \text{Não é LI (L.D.)}$

13-

a) $x\vec{u} - y\vec{v} = (2x - y, -1x + y)$

$$\begin{cases} 2x - y = 1 \\ -x + y = 1 \end{cases} \quad \begin{matrix} + & -2 + y = 1 \\ & y = 3 \end{matrix} \quad \therefore 2\vec{u} - 3\vec{v} = \vec{w}$$

$$x = 2 \quad y = 3$$

b)

$$\begin{cases} x + z = 1 & z = 1 - x & x + z = 1 \\ x + y + z = 2 & \Rightarrow x + y + 1 - x = 2 & 2 + z = 1 \\ x + y = 3 & y = 1 & z = -1 \\ x + 1 = 3 & & \\ x = 2 & \vec{z} = 2\vec{a} + \vec{b} - \vec{c} \end{cases}$$

14-

$$\begin{cases} 1 = K \cdot m & \rightarrow \left(\frac{m}{9}\right), m = 1 \rightarrow m^2 = 9 \rightarrow m = \pm 3 \\ m - 1 = K \cdot 2n & \text{Se } m = 2 \\ m = K \cdot 9 & \Rightarrow K = \frac{m}{9} \quad K = \frac{2}{9} = \frac{1}{2} \end{cases}$$

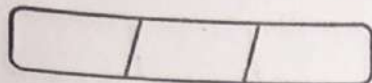
Para $m = 2$ e $n = 1$ $2 - 1 = \frac{1}{2} \cdot 2n$
 $\vec{u}(1, 1, 2) \quad \vec{v}(2, 2, 4) \rightarrow \vec{u} = \frac{1}{2}\vec{v} \quad n = 1$

Se $m = -2$ $K = -\frac{1}{2}$
 Para $m = -2$ e $n = 3$ $n = 3$

$\vec{u}(1, -3, -2) \quad \vec{v}(-2, 6, 4) \rightarrow \vec{u} = -\frac{1}{2}\vec{v}$

$\therefore (m, n) = (2, 1) \text{ e } (-2, 3)$





$$b) \begin{cases} 1 = km = k = \frac{1}{m} \\ m = k(n+1) \Rightarrow m = \frac{1}{m}(n+1) = m^2 = n+1 \\ n+1 = k \cdot 8 \Rightarrow m^2 = 8 = m^3 = 8 \\ \quad \quad \quad m \quad \quad \quad m=2 \end{cases}$$

$$2 = \frac{1}{2} \cdot (n+1)$$

$$n+1 = 4$$

$$n = 3$$

Para $m=2$ e $n=3$

$$\vec{u} = (1, 2, 4) \quad \vec{v} = (2, 4, 8) = \vec{u} = \frac{1}{2} \vec{v}$$

$\therefore$ Os valores são $(m, n) = (2, 3)$

$$15- \begin{vmatrix} m & -1 & m^2+1 \\ m^2+1 & m & 0 \\ m & 1 & 1 \end{vmatrix} = m^2 + (m^2+1)(m^2+1) + 0 - m^4 - m^2 + m^2+1 \\ m^2 + m^4 + m^2 + m^2 + m^4 - m^4 - m^2 + m^2+1 \\ 1 m^2 + 2 m^4 - m^2 - 2 m^4 + 2 \neq 0$$

E.L.I. independente de

$$3m^2 + 2 \neq 0$$

Valor de m

Para todo $m \in \mathbb{R} \Rightarrow \text{Det} \neq 0; 2 \neq 0$

$$16-a) C = (1, 1, 0) (1, 0, 1) (1, 1, -1)$$

$$\begin{vmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 1 & 1 & -1 \end{vmatrix} = 0 + 0 + 1 - 0 - 1 + 1 \\ 1 \neq 0 \text{ logo L.I., entao e um plano}$$

$$\vec{u} = (2, 3, 7)$$

$$b) \vec{u} = 2\vec{p}_1 + 3\vec{p}_2 + 7\vec{p}_3$$

$$\vec{u} = 2(1, 1, 0) + 3(1, 0, 1) + 7(1, 1, -1)$$

$$\vec{u} = (2+3+7, 2+0+7, 0+3-7)$$

$$\vec{u} = (12, 9, -4)$$

$$c) \vec{N} = (2, 3, 7) = c_1(1, 1, 0) + c_2(1, 0, 1) + c_3(1, 1, -1)$$

$$\begin{cases} c_1 + c_2 + c_3 = 2 \Rightarrow 3 - c_3 + c_2 + c_3 = 2 \Rightarrow c_2 = -1 \\ c_1 + c_3 = 3 \Rightarrow c_1 = 3 - c_3 \Rightarrow c_1 = 3 + 8 = 11 \\ c_2 + c_3 = 7 \end{cases}$$

$$\vec{N} = (11, -1, -8)$$

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$$\begin{aligned} -1 - c_3 &= 7 \\ c_3 &= -8 \end{aligned}$$

